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# 1 Fundamentals of Semiconductor Devices

Reference basis: NPTEL course - Fundamentals of Semiconductor Devices By Prof. Digbijoy N. Nath | IISc Bangalore

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## 1.1 Introduction to semiconductors

- Semiconductors are materials whose electrical conductivity lies between conductors and insulators and can be controlled using doping, temperature, and electric fields.
- They form the foundation of modern electronic devices such as diodes, transistors, and integrated circuits.
- Conduction occurs via two types of charge carriers: **electrons (negative)** and **holes (positive)**.
- Silicon (Si) is the most widely used semiconductor due to its stability and ease of fabrication.
- Their behavior is governed by quantum mechanics and band theory.

**Moore's Law** states that the number of transistors on an integrated circuit doubles approximately every 18–24 months (~1.5-2 years), leading to exponential growth in computing power. - This scaling results in higher performance, lower cost per transistor, and increased device density. - However, physical and thermal limits are slowing down this trend in modern technologies.

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## 1.2 Energy bands

- In crystals, atomic energy levels combine to form continuous bands due to interaction between atoms.
  - Energy Bands are ranges of allowed electron energies formed due to atomic orbital overlap in a crystal lattice.
  - Electrons occupy energy bands: **valence band (VB)** and **conduction band (CB)**.
  - Valence Band: Electron rich bands, holds bound electrons
  - Conduction Band: Electron deficient bands, allows free electron movement
  - The **bandgap (E<sub>g</sub>)** is the energy difference between VB and CB.
  - Small bandgap allows electrons to move to CB and conduct electricity.
  - Insulators have large bandgap, while conductors have overlapping bands. Semiconductors have Moderate & achievable bandgap.
  - Band structure determines electrical and optical properties.
  - Holes: Absence of electrons
  - Fermi level: Shows probability of finding electronics along the energy band gap
- 

## 1.3 Types of Semiconductors

- **Intrinsic Semiconductors:** Pure materials with equal electron and hole concentration ( $n = p = n_i$ ).
- Intrinsic semiconductor (extremely pure): Materials which have exactly same holes and electrons or Fermi level sits exactly in the middle
- **Extrinsic Semiconductors:** Doped materials with controlled carrier concentration.

### 1.3.1 Doping

- **Doping** is a process of adding impurities (either holes or electrons) to the intrinsic semiconductor to make it either n-type or p-type semiconductor
  - This process is inherent only to the semiconductors. This cannot be done to conductors or insulators.
    - Eg. Adding Boron or Phosphorus can be added to the Si lattice
  - Doping introduces donors (n-type) or acceptors (p-type). Based on these, we get 2 types of Extrinsic Semiconductors:
    - **n-type semiconductor:** Materials which have more electrons than holes or Fermi level is in the upper half
      - electrons are majority carriers in n-type semiconductors.
    - **p-type semiconductor:** Materials which have more holes than electrons or Fermi level is in the lower half
      - holes are majority carriers in p-type semiconductors.
  - Doping enables control over conductivity.
  - Doping enables control of electrical properties.
  - Small doping significantly increases conductivity.
  - Intrinsic carrier concentration depends on temperature and bandgap.
-

## 1.4 Band Structure

### 1.4.1 Band structure & fermi-dirac distribution

- Fermi-Dirac distribution gives the probability of an energy state being occupied:

$$f(E) = \frac{1}{1 + e^{(E-E_F)/kT}}$$

- At  $T = 0K$ , all states below  $E_F$  are filled and above are empty (step-like distribution).
- At finite temperature, electrons get thermally excited to higher energy states, smoothing the distribution.
- Fermi level shifts with doping: toward conduction band (n-type) and toward valence band (p-type).
- It directly determines electron and hole concentrations in semiconductors.

### 1.4.2 Density of states (DOS)

- Density of states (DOS) represents the number of available energy states per unit energy per unit volume.
- In 3D semiconductors, DOS increases as

$$\sqrt{E - E_c}$$

above the conduction band edge.

- Carrier concentration is obtained by combining DOS with the Fermi-Dirac distribution.
- DOS depends on effective mass and dimensionality of the system.
- It is essential for calculating electron and hole populations in materials.

### 1.4.3 Equilibrium carrier concentration

- At thermal equilibrium, carrier concentrations follow:

$$np = n_i^2$$

- Charge neutrality ensures total positive and negative charges are balanced.
- Majority carriers are approximately equal to dopant concentration.
- Minority carriers are derived from intrinsic relations.
- Fermi level position uniquely determines equilibrium carrier densities.

### 1.4.4 Temperature dependence

- Intrinsic carrier concentration increases exponentially with temperature:

$$n_i \propto e^{-E_g/2kT}$$

- Three regions: freeze-out (low T), extrinsic (moderate T), intrinsic (high T).
- At high temperatures, intrinsic carriers dominate over dopants.
- Mobility decreases with temperature due to increased phonon scattering.
- Temperature strongly impacts device behavior and reliability.

#### 1.4.5 High doping & incomplete ionization

- Heavy doping leads to **degenerate semiconductors**, where Fermi level enters a band.
- Bandgap narrowing occurs due to interaction between closely spaced dopants.
- At low temperatures, dopants may not fully ionize (incomplete ionization).
- Carrier concentration deviates from simple doping assumptions.
- Critical in heavily doped regions like MOSFET source/drain.

#### 1.4.6 Carrier scattering & mobility

- Mobility defines how easily carriers move under an electric field:

$$\mu = \frac{q\tau}{m^*}$$

- Scattering mechanisms include phonon (lattice) scattering and impurity scattering.
- Higher temperature increases phonon scattering, reducing mobility.
- Higher doping increases impurity scattering, also reducing mobility.
- Mobility directly affects conductivity and switching speed of devices.

#### 1.4.7 Low & high field transport

- At low electric fields, carrier velocity is proportional to electric field (Ohm's law).
- At high fields, velocity saturates due to increased scattering.
- Velocity saturation limits current in short-channel devices.
- High-field effects cause deviation from linear transport models.
- Important for modern nanoscale transistor design.

#### 1.4.8 Drift-diffusion & traps

- Drift current arises due to electric field:

$$J_{drift} = qn\mu E$$

- Diffusion current arises due to carrier concentration gradient:

$$J_{diff} = qD \frac{dn}{dx}$$

- Einstein relation links diffusion and mobility:

$$\frac{D}{\mu} = \frac{kT}{q}$$

- Trap states capture carriers and influence recombination and leakage.
- Traps introduce non-ideal behavior such as reduced carrier lifetime.

#### 1.4.9 Charge transport

- Total current is the sum of drift and diffusion currents.
- Drift dominates under strong electric fields; diffusion dominates under concentration gradients.
- Both electrons and holes contribute to current flow.
- Carrier mobility determines how quickly carriers respond to applied fields.
- These mechanisms form the basis of all semiconductor device operation.

### 1.4.10 Continuity equation

- Ensures conservation of charge:

$$\frac{\partial n}{\partial t} = G - R + \frac{1}{q} \nabla J$$

- Accounts for generation (G), recombination (R), and current flow (J).
  - Used for both steady-state and transient analysis of devices.
  - Fundamental equation in semiconductor device modeling and simulation.
  - Connects microscopic carrier dynamics to measurable electrical current.
- 

## 1.5 PN Junction

- A PN junction is formed by joining **p-type (hole-rich)** and **n-type (electron-rich)** semiconductor regions.
- Due to carrier concentration difference, electrons (from n-side) and holes (from p-side) **diffuse across the junction**.
- This diffusion leaves behind fixed ionized charges, forming a **depletion region** with no mobile carriers.
- An internal **electric field (built-in potential)** is created, which opposes further diffusion.
- At equilibrium, drift and diffusion currents balance, resulting in **zero net current**.
- The junction has a field and prevents the current from flowing with no field applied across the semiconductors.

### 1.5.1 Biasing of PN Junction

#### 1. Forward Bias

- p-side connected to positive terminal, n-side to negative terminal.
- Reduces barrier potential and narrows depletion region.
- Allows majority carriers to cross the junction easily.
- Results in **large exponential current flow**:

$$I = I_s(e^{V/V_T} - 1)$$

- Turn-on voltage (~0.7V for Si) is required to significantly conduct.

#### 2. Reverse Bias

- p-side connected to negative terminal, n-side to positive terminal.
- Increases barrier potential and widens depletion region.
- Prevents majority carrier flow.
- Only small **reverse saturation current** flows due to minority carriers.
- Acts like an insulator under normal conditions.

### 1.5.2 Depletion Region & Built-in Potential

- The depletion region acts as a **barrier** preventing free carrier movement.

- Built-in potential  $V_{bi}$  depends on doping levels:

$$V_{bi} \propto \ln \left( \frac{N_A N_D}{n_i^2} \right)$$

- Width of depletion region changes with applied voltage.
- This region is crucial for controlling current flow in the device.
- Acts like an insulating layer between p and n regions.
- When voltage is applied, it takes some voltage to overcome the depletion region before current starts flowing. This is called as turn on voltage.
- The semiconductor becomes operational when more voltage is applied than turn on voltage.

### 1.5.3 Breakdown Mechanisms

- At high reverse voltage, junction undergoes **breakdown**, causing large current.

### 1.5.4 Types:

- **Zener (Tunneling) Breakdown**
    - Occurs in heavily doped junctions at low voltage.
    - Strong electric field enables quantum tunneling.
  - **Avalanche Breakdown**
    - Occurs in lightly doped junctions at high voltage.
    - Carriers gain energy and generate more carriers via collisions.
  - Breakdown can damage the device unless properly controlled (e.g., Zener diodes).
- 

## 1.6 Applications of PN Junction

### 1.6.1 Rectifier (Diode)

- Converts AC to DC by allowing current in one direction only.
- When connected in Forward Bias, the junction allows the current to flow.
- In Reverse Bias, prevents current flow except for a small leakage current.
- Used in power supplies and electronic circuits.

### 1.6.2 LED (Light Emitting Diode)

- An LED (Light Emitting Diode) is a PN junction diode that emits light when forward biased.
- Electrons and holes recombine at the junction, releasing energy as photons (light).
- Emits light when forward biased due to **radiative recombination**.
- Electron-hole recombination releases photons:

$$E_{\text{photon}} \approx E_g$$

- Color depends on semiconductor bandgap.
- Used in displays, lighting, and indicators.

### 1.6.3 Photodetector (Photodiode)

- A photodetector is a PN junction device that converts light into electrical current.
- When light photons strike the junction, they generate electron-hole pairs in the depletion region.
- Under reverse bias, these carriers are quickly swept across the junction, producing photocurrent.
- Also can be used in any spectrum: IR, UV, Infrared etc.
- Used in optical communication and sensors.
- 

### 1.6.4 Solar Cell

- A solar cell is a PN junction device that converts sunlight directly into electrical energy using photovoltaic effect.
  - Photons striking the junction create electron-hole pairs, generating current.
  - It operates under illumination without external bias, using built-in potential to separate charges.
  - Only sunlight is the input spectra.
  - Used in renewable energy systems.
- 

## 1.7 Schottky Junction (Metal–Semiconductor Junction)

- A Schottky junction is formed when a **metal is brought into contact with a semiconductor** (typically n-type).
- Instead of a p–n interface, it creates a **metal–semiconductor interface** with unique electrical behavior.
- Charge transfer occurs at the interface, leading to formation of a **Schottky barrier**.
- It is also called a **Schottky diode** when used as a device.

### 1.7.1 Formation & Barrier

- When metal and semiconductor come in contact, electrons flow to align their Fermi levels.
- This creates a **depletion region in the semiconductor side only** (not in the metal).
- A potential barrier called the **Schottky barrier height** ( $\phi_B$ ) is formed.
- Barrier height depends on **metal work function and semiconductor properties**.
- The built-in electric field prevents further carrier diffusion at equilibrium.

### 1.7.2 Operation

#### 1. Forward Bias

- Reduces the barrier height at the interface.
- Majority carriers (electrons in n-type) easily cross into the metal.



- Results in **low forward voltage drop** ( $\sim 0.2\text{--}0.3\text{ V}$ ) compared to silicon PN diodes ( $\sim 0.7\text{ V}$ ).
- Current increases rapidly with applied voltage.

## 2. Reverse Bias

- Increases barrier height and widens depletion region.
- Prevents majority carrier flow.
- Only small leakage current flows (higher than PN junction).
- No significant charge storage occurs.

### 1.7.3 Key Characteristics

- **Majority carrier device**  $\rightarrow$  no minority carrier storage.
- **Fast switching speed** due to absence of charge storage.
- **Low forward voltage drop**  $\rightarrow$  lower conduction losses.
- **Higher reverse leakage current** compared to PN junction.
- Barrier properties depend on **choice of metal**.

### 1.7.4 Advantages

- High-speed operation (ideal for RF and switching circuits).
- Low power loss due to low forward voltage.
- Simple structure compared to PN junction.

### 1.7.5 Limitations

- Higher reverse leakage current.
- Lower breakdown voltage compared to PN diodes.
- Sensitive to temperature variations.

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## 1.8 Transistors

- A transistor is a **three-terminal semiconductor device** used to control current flow and signal behavior.
- It operates by using a **small input signal** to control a much larger current between two terminals.
- The three terminals depend on type of transistor:
  - BJT: Emitter, Base, Collector
  - FET: Source, Gate, Drain
- Transistors are the **fundamental building blocks** of modern electronic circuits.

### 1.8.1 Key Functions / Applications

- **Switch:** Turns current ON/OFF (used in digital circuits).
- **Amplifier:** Increases signal strength (analog circuits).
- **Digital Logic:** Forms logic gates and processors (CMOS technology).
- **Sensors:** Used in sensing circuits and signal conditioning.
- **Memory:** Stores data in SRAM, DRAM, and Flash technologies.

### 1.8.2 Types of Transistors

#### 1. Field Effect Transistor (FET)

- A **voltage-controlled device** where current is controlled by an electric field at the gate.
- Terminals: **Source, Gate, Drain**.
- Gate voltage controls the conductivity of the channel between source and drain.
- Very high input impedance (almost no gate current).
- Examples: **MOSFET, JFET** → widely used in digital electronics.

#### 2. Bipolar Junction Transistor (BJT)

- A **current-controlled device** that uses both electrons and holes (bipolar conduction).
  - Terminals: **Emitter, Base, Collector**.
  - A small base current controls a large collector current (current amplification).
  - Types: **NPN and PNP transistors**.
  - Used in amplification and analog circuits.
- 

## 1.9 Bipolar Junction Transistor

- A BJT (Bipolar Junction Transistor) is a three-terminal semiconductor device (emitter, base, collector) that can amplify or switch signals.
- It uses both electrons and holes as charge carriers, hence the term “bipolar”.
- **Base (B)** Thin, lightly doped region that controls carrier flow from emitter to collector by regulating recombination.
- **Emitter (E)** Heavily doped region that injects majority carriers into the base.
- **Collector (C)** Moderately doped region that collects carriers from the base and delivers output current.
- **Currents from E, B, C** : Emitter current splits into base and collector currents:  $I_E = I_B + I_C$
- **Gain (Current Gain)** : Ability of BJT to amplify current, defined as ratio of collector current to base current:

$$\beta = \frac{I_C}{I_B}$$

- **Beta ( )** : Common-emitter current gain representing amplification capability of the transistor:

$$\beta = \frac{I_C}{I_B}$$

- **Base Transport Factor ( )** : Fraction of carriers injected from emitter that successfully reach the collector through the base:

$$\alpha_T = \frac{I_C}{I_E}$$

- **Emitter Injection Efficiency (  $\gamma$  )** : Fraction of emitter current that contributes to useful carrier injection:

$$\gamma = \frac{\text{electron injection}}{\text{total emitter current}}$$

- **Transport Ratio / Common-base Gain (  $\alpha$  )** : Fraction of emitter current that becomes collector current:

$$\alpha = \frac{I_C}{I_E}$$

- **Important Relation :**

$$\beta = \frac{\alpha}{1 - \alpha}$$

### 1.9.1 BJT Operation

- When a small positive voltage is applied to the base relative to the - emitter, electrons are injected from the emitter into the base.
- These electrons cross the thin base region and are collected by the - collector, creating a large collector current.
- Thus, a small base current controls a much larger collector-emitter current, enabling amplification or switching.
- Emmitter-Base junction is placed in forward bias & Base-Collector is placed in reverse bias.
- A small current at the base controls a much larger current between collector and emitter.
- For best BJT operation & improving gain, Beta & other parameters, base should be narrow & lightly doped.
- BJTs has 2 types:
  1. NPN: An NPN transistor has an n-type emitter, p-type base, and n-type - collector.
  2. PNP: An PNP transistor has an p-type emitter, n-type base, and p-type - collector.

## 1.10 MOSFET

### 1.10.1 MOS (Metal-oxide-semiconductor) CAPACITOR

- A MOS capacitor consists of a Metal–Oxide–Semiconductor structure where the oxide acts as an insulator and the gate voltage controls charge at the semiconductor surface.
- Semiconductor can be p-type or n-type doped.
- Applying a gate voltage changes the surface potential, causing redistribution of carriers near the oxide-semiconductor interface.
- Depending on gate voltage polarity and magnitude, the device operates in accumulation, depletion, or inversion.
- No current flows through the oxide (ideal case); only charge rearrangement occurs.
- MOS capacitor behavior is fundamental to understanding MOSFET operation.

- Accumulation: Occurs when gate voltage attracts majority carriers to the surface, increasing carrier concentration near the interface.
- Depletion: Occurs when gate voltage repels majority carriers, leaving behind fixed ionized charges and forming a depletion region.
- Inversion: Occurs when sufficient gate voltage attracts minority carriers, forming a conductive channel at the surface.
- Threshold Voltage  $V_T$ : The gate voltage at which strong inversion begins and a conducting channel is formed.

### 1.10.2 MOSFET definition

- A MOSFET (Metal–Oxide–Semiconductor Field-Effect Transistor) is a three-terminal device (Gate, Source, Drain) where current between source and drain is controlled by gate voltage.
- It is derived from the MOS capacitor by adding two heavily doped regions (source and drain) in the semiconductor.
- The gate voltage controls the formation of a conductive channel at the semiconductor surface.
- It is a voltage-controlled device with very high input impedance due to the insulating oxide.
- MOSFETs are the fundamental building blocks of modern digital and analog circuits.

### 1.10.3 Relation to MOS Capacitor

- The MOSFET structure is essentially a MOS capacitor with source and drain added for current flow.
- The same surface charge control principles (accumulation, depletion, inversion) apply.
- Channel formation in MOSFET occurs under **inversion condition** of MOS capacitor.
- Threshold voltage of MOS capacitor directly determines when MOSFET turns ON.
- Thus, MOSFET operation is an extension of MOS capacitor electrostatics.

### 1.10.4 Operation of MOSFET (nMOS example)

#### 1. Cutoff Region (OFF)

- When  $V_{GS} < V_T$ , no inversion layer forms and no conduction path exists.
- Only very small leakage current flows.

#### 2. Linear / Triode Region

- When  $V_{GS} > V_T$  and  $V_{DS}$  is small, a continuous channel forms.
- Device behaves like a voltage-controlled resistor.
- Drain current:

$$I_D = \mu_n C_{ox} \frac{W}{L} \left[ (V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

#### 3. Saturation Region

- When  $V_{DS} = (V_{GS} - V_T)$ , channel pinches off near drain.
- Current becomes almost independent of  $V_{DS}$ .
- Drain current:

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_T)^2$$

### 1.10.5 Key Physical Insight

- Gate voltage controls surface potential → creates inversion layer → forms channel → enables current flow.
- MOSFET current is controlled purely by electric field (no gate current ideally).
- Device transitions from **no channel** → **resistive channel** → **pinch-off region** as voltages change.

### 1.10.6 Important Parameters

- $V_T$ : Threshold voltage (onset of inversion)

•

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

: Oxide capacitance per unit area

- $\mu_n$ : Carrier mobility
- $W/L$ : Device geometry (width/length ratio)
- Gate voltage controls channel formation.
- Regions: cutoff, linear, saturation.
- Current depends on gate voltage.
- Short-channel effects degrade performance.
- Most important device in digital electronics.

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## 1.11 Compound Semiconductors

- Compound semiconductors are materials formed by combining two or more elements (typically from III–V or II–VI groups) such as GaAs, InP, and GaN.
- They offer superior electronic and optical properties compared to silicon, especially for high-speed and optoelectronic applications.
- Many compound semiconductors have a **direct bandgap**, enabling efficient light emission and absorption.
- Their properties can be engineered by changing composition (e.g., AlGaAs, InGaN).
- Widely used in LEDs, lasers, RF devices, and high-frequency electronics.
- Compound semiconductors enable **high-speed + optical + high-power applications** where silicon falls short.

| Group / Material Type  | Example Materials | Key Properties                                                     | Typical Applications                               |
|------------------------|-------------------|--------------------------------------------------------------------|----------------------------------------------------|
| <b>III–V Arsenides</b> | GaAs, AlGaAs      | Direct bandgap, high electron mobility, moderate bandgap (~1.4 eV) | RF devices, microwave circuits, LEDs, laser diodes |

| Group / Material Type                | Example Materials       | Key Properties                                                 | Typical Applications                                  |
|--------------------------------------|-------------------------|----------------------------------------------------------------|-------------------------------------------------------|
| <b>III–V Phosphides</b>              | InP, GaP                | High electron velocity, good thermal stability                 | High-speed electronics, optical communication, lasers |
| <b>III–V Nitrides</b>                | GaN, AlGaN, InGaN       | Wide bandgap, high breakdown field, high temperature stability | Power electronics, LEDs (blue/white), RF amplifiers   |
| <b>III–V Antimonides</b>             | InSb, GaSb              | Very narrow bandgap, very high mobility                        | Infrared detectors, thermal imaging, sensors          |
| <b>II–VI Semiconductors</b>          | CdS, CdSe, ZnSe         | Direct bandgap, strong optical interaction                     | Photodetectors, LEDs, display technologies            |
| <b>Ternary Compounds</b>             | AlGaAs, InGaN           | Tunable bandgap via composition                                | LEDs, laser diodes, heterostructure devices           |
| <b>Quaternary Compounds</b>          | InGaAsP, AlGaInP        | Precise bandgap engineering, lattice matching                  | Fiber-optic communication, high-efficiency lasers     |
| <b>Heterostructures</b>              | GaAs/AlGaAs, InP/InGaAs | Band offset engineering, carrier confinement                   | HEMTs, HBTs, high-speed transistors                   |
| <b>Wide Bandgap Semiconductors</b>   | GaN, SiC                | High breakdown voltage, high temperature operation             | Power devices, EV electronics, high-frequency systems |
| <b>Narrow Bandgap Semiconductors</b> | InSb, HgCdTe            | Sensitive to IR radiation, low energy gap                      | Infrared photodetectors, night vision systems         |

- **Direct bandgap** → optoelectronics (**LEDs, lasers**)
- **High mobility** → high-speed RF devices
- **Wide bandgap** → power electronics
- **Narrow bandgap** → infrared detection

#### 1.11.1 Key Properties

- **High electron mobility** → faster carrier transport and high-speed device operation.
- **Direct bandgap (in many materials)** → efficient light emission (used in LEDs and lasers).
- **Wide bandgap (e.g., GaN)** → high power and high-temperature operation.
- **Bandgap engineering** → allows tuning of optical and electrical properties.
- **Heterostructure compatibility** → enables advanced device design.

#### 1.11.2 Comparison with Silicon

- Compound semiconductors have higher mobility and better optical properties than silicon.

- Silicon has indirect bandgap  $\rightarrow$  poor light emission, while many compounds are direct bandgap.
- Compound semiconductors are more expensive and harder to fabricate.
- Silicon dominates digital ICs, while compound semiconductors dominate optoelectronics and RF.
- Trade-off exists between cost, performance, and manufacturability.

### 1.11.3 Applications

- **Optoelectronics:** LEDs, laser diodes (GaAs, InGaN).
  - **High-frequency devices:** RF amplifiers, microwave circuits (GaAs, InP).
  - **Power electronics:** GaN-based high-voltage devices.
  - **Solar cells:** High-efficiency multi-junction cells.
  - **Sensors & photodetectors:** Fast optical detection systems.
- 

## 1.12 Solar Cells

- A solar cell is a semiconductor device that converts solar light energy into electrical energy using the photovoltaic effect.
- It is typically a **p–n junction device**, where incident photons generate electron–hole pairs.
- The built-in electric field in the depletion region separates these carriers, producing current.
- When connected to a load, this current flows as usable electrical power.
- Solar cells are the fundamental units of solar panels used in renewable energy systems.
- **Solar cells do not require external voltage application for generating current.**
- **Short-Circuit Current ( $I_{sc}$ ) :** The current flowing through the solar cell when the output terminals are shorted (i.e.,  $V = 0$ ), representing the maximum current generated under illumination.
- **Open-Circuit Voltage ( $V_{oc}$ ) :** The voltage across the solar cell when no external current flows (i.e.,  $I = 0$ ), representing the maximum voltage developed due to carrier separation.

### 1.12.1 Working Principle

- Photons with energy greater than bandgap ( $h\nu > E_g$ ) generate electron–hole pairs.
- The internal electric field drives electrons toward the n-side and holes toward the p-side.
- This separation prevents recombination and creates a voltage across the junction.
- External circuit allows current to flow, delivering power.
- Continuous illumination maintains carrier generation and output.

### 1.12.2 Loss Mechanisms

- Recombination of carriers before collection reduces efficiency.
- Reflection losses prevent light from entering the device.
- Resistive losses in contacts and material reduce output power.
- Thermalization losses occur when excess photon energy is lost as heat.

- Imperfect absorption limits carrier generation.

### 1.12.3 Types of Solar Cells

- **Silicon Solar Cells:** Most common, reliable, moderate efficiency.
  - **Thin-Film Solar Cells:** Lightweight, flexible, lower efficiency.
  - **Compound Semiconductor Cells:** High efficiency (GaAs, multi-junction).
  - **Perovskite Solar Cells:** Emerging, high efficiency potential.
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## 1.13 Photodetectors

- A photodetector is a semiconductor device that converts incident light with a certain wavelength (Infrared, visible, Ultraviolet etc.) into an electrical signal by generating electron–hole pairs.
- It typically operates using a **p–n or p–i–n junction**, where light absorption creates carriers.
- The built-in or applied electric field separates these carriers, producing current.
- Output signal is proportional to incident optical power.
- Widely used in optical communication, sensing, and imaging systems.

### 1.13.1 Working Principle

- Incident photons with energy  $h\nu \geq E_g$  generate electron–hole pairs.
- These carriers are separated by the electric field in the depletion region.
- Electrons move toward n-side and holes toward p-side, creating photocurrent.
- Reverse bias is often applied to increase depletion width and speed.
- Faster response is achieved with thinner active regions and strong fields.

### 1.13.2 Types of Photodetectors

- **PN Photodiode:** Basic structure, moderate speed and sensitivity.
- **PIN Photodiode:** Wider depletion region give better sensitivity and faster response.
- **Avalanche Photodiode (APD):** Internal gain via impact ionization  $\rightarrow$  high sensitivity.
- **Photoconductor:** Conductivity changes with light, slower but simple.

### 1.13.3 Key Performance Parameters

- **Responsivity (R):** Output current per unit optical power:

$$R = \frac{I_{ph}}{P_{opt}}$$

- **Quantum Efficiency ( $\eta$ ):** Fraction of photons converted into carriers.
- **Dark Current:** Leakage current in absence of light.
- **Response Time / Bandwidth:** Speed of detector operation.
- **Noise:** Limits sensitivity and minimum detectable signal.



### 1.13.4 Important Concepts

- Higher absorption coefficient results in more carrier generation.
- Larger depletion region yield in higher efficiency but slower response (trade-off).
- Reverse bias improves speed but increases noise.
- Avalanche multiplication enhances signal but adds excess noise.
- Material bandgap determines wavelength sensitivity.

#### 1.13.4.1 Photodetectors vs Solar Cells

| Feature                    | Photodetectors                                            | Solar Cells                                                          |
|----------------------------|-----------------------------------------------------------|----------------------------------------------------------------------|
| <b>Purpose</b>             | Detect light and convert it into an electrical signal     | Convert light into usable electrical power                           |
| <b>Primary Goal</b>        | Sensitivity and speed                                     | Maximum energy conversion efficiency                                 |
| <b>Operation Mode</b>      | Usually operated in <b>reverse bias</b> for fast response | Operated in <b>zero bias or forward bias</b> (power generation mode) |
| <b>Output</b>              | Small signal current proportional to light intensity      | Continuous power output (voltage + current)                          |
| <b>Speed</b>               | Very fast (ns-ps range for high-speed devices)            | Relatively slow (not optimized for speed)                            |
| <b>Responsivity</b>        | High sensitivity to small optical signals                 | Designed for maximum total energy capture                            |
| <b>Efficiency Focus</b>    | Detection efficiency (quantum efficiency)                 | Power conversion efficiency ( )                                      |
| <b>Structure</b>           | PN, PIN, Avalanche photodiodes                            | Mainly PN junction (sometimes multi-junction)                        |
| <b>Gain Mechanism</b>      | Can include internal gain (APD via avalanche)             | No internal gain mechanism                                           |
| <b>Noise Consideration</b> | Noise is critical (limits detection capability)           | Less critical compared to power output                               |
| <b>Applications</b>        | Optical communication, sensors, imaging                   | Solar panels, renewable energy systems                               |

### 1.14 Recombination in LEDs

- Recombination is the process where electrons from the conduction band combine with holes in the valence band, releasing energy in the form of photons (light).
- This is primarily **radiative recombination**, which is the desired mechanism for light emission.
- Occurs in the active region (usually a direct bandgap semiconductor).
- Efficiency of LED depends on how many recombinations produce photons.
- Direct bandgap materials (e.g., GaN, GaAs) are used to maximize light emission.

### 1.14.1 Types of Recombination

- **Radiative Recombination:** Electron-hole recombination that emits a photon (useful for LEDs).
- **Non-radiative (SRH) Recombination:** Energy lost as heat via trap states, reducing efficiency.
- **Auger Recombination:** Energy transferred to another carrier instead of emitting light, dominant at high carrier densities.

### 1.14.2 Key Concepts

- Radiative recombination rate is proportional to electron and hole concentrations:

$$R_{rad} \propto np$$

- Internal Quantum Efficiency (IQE):

$$IQE = \frac{\text{radiative recombination}}{\text{total recombination}}$$

- Higher radiative recombination results in brighter LED output.
- Non-radiative processes reduce efficiency and generate heat.
- Carrier confinement (using heterostructures) improves recombination efficiency.

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## 1.15 LED (Light Emitting Diode)

- An LED is a semiconductor **p–n junction device** that emits light when forward biased due to electron–hole recombination.
- It uses **direct bandgap materials** (e.g., GaN, GaAs) to efficiently convert electrical energy into light.
- The emitted light wavelength (color) depends on the bandgap of the material.
- LEDs are widely used in displays, lighting, and indicators due to high efficiency and long life.
- It is an optoelectronic device combining electrical and optical phenomena.

### 1.15.1 Operation Principle

- When forward bias is applied, electrons from n-side and holes from p-side are injected into the junction.
- These carriers recombine in the active region.
- In direct bandgap materials, recombination releases energy as photons (light emission).
- Photon energy is approximately equal to bandgap:

$$E_{photon} \approx E_g = h\nu$$

- Continuous carrier injection results in continuous light output.

### 1.15.2 Key Concepts

- **Forward Bias** is essential for carrier injection and light emission.
- **Radiative recombination** produces light, while non-radiative processes reduce efficiency.
- **Internal Quantum Efficiency (IQE)** measures how efficiently carriers produce photons.
- **External Quantum Efficiency (EQE)** also includes light extraction efficiency.
- Device design focuses on maximizing light output and minimizing losses.

### 1.15.3 Important Factors Affecting LED Performance

- Material choice (direct bandgap required).
  - Carrier recombination efficiency.
  - Light extraction efficiency (internal reflections reduce output).
  - Temperature (higher temperature reduces efficiency).
  - Device structure (heterostructures improve confinement).
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## 1.16 Advanced transistors

### 1.16.1 Transistors for Power Electronics

- Power transistors are semiconductor devices designed to handle **high voltage, high current, and high power levels** efficiently.
- They are used as switches or amplifiers in power conversion systems.
- Unlike small-signal devices, they are optimized for **low losses and high reliability**.
- Common types include **Power MOSFETs, IGBTs, and BJTs**.
- Widely used in power supplies, inverters, motor drives, and EV systems.

#### 1.16.1.1 Key Requirements

- **High breakdown voltage** to withstand large voltages in OFF state.
- **Low ON-state resistance** to reduce conduction losses.
- **High current handling capability** for power applications.
- **Efficient heat dissipation** to manage thermal effects.
- **Fast switching capability** to minimize switching losses.

#### 1.16.1.2 Operation Principle

- Power transistors operate mainly as **switches** (ON/OFF states).
- In ON state low resistance path allows current flow.
- In OFF state high resistance blocks current.
- Switching between states enables energy control and conversion.
- Efficiency depends on minimizing conduction and switching losses.

#### 1.16.1.3 Types of Power Transistors

- **Power MOSFET**: Fast switching, low gate power, used in low–medium voltage.
- **IGBT (Insulated Gate Bipolar Transistor)**: Combines MOSFET control with BJT conduction, used in high power.
- **Power BJT**: High current capability but slower switching, less commonly used now.

#### 1.16.1.4 Loss Mechanisms

- **Conduction Loss:** Due to ON resistance or voltage drop.
- **Switching Loss:** During transitions between ON and OFF states.
- **Leakage Loss:** Small current in OFF state.
- **Thermal Loss:** Heat generated affects efficiency and reliability.

#### 1.16.2 Transistors for Memory

##### 1.16.2.1 What are Memory Transistors?

- Memory transistors are semiconductor devices designed to **store data (bits) by retaining charge or state**.
- They form the basic building blocks of memory technologies like **SRAM, DRAM, and Flash**.
- Unlike logic transistors, they are optimized for **data retention, stability, and low leakage**.
- Information is stored as either charge, voltage level, or stored state.
- Used in caches, RAM, and non-volatile storage systems.

##### 1.16.2.2 Operation Principle

- Memory operation involves **write, store, and read** cycles.
- During write, charge is stored in a node or capacitor.
- During hold, the device retains stored information for a certain duration.
- During read, stored data is sensed without disturbing it significantly.
- Data is represented as binary states (0 or 1).

##### 1.16.2.3 Types of Memory Transistor Structures

- **SRAM (Static RAM):** Uses multiple transistors (typically 6T) to store a stable state using feedback.
  - **DRAM (Dynamic RAM):** Uses one transistor + capacitor; requires periodic refresh.
  - **Flash Memory:** Uses floating gate MOSFET to store charge permanently (non-volatile).
  - **EEPROM:** Allows electrical erasing and reprogramming.
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### 1.17 Microelectronic Fabrication

- Microelectronic fabrication is the process of creating semiconductor devices and integrated circuits on a wafer (typically silicon).
- It involves a sequence of physical and chemical processes to build structures at micro/nano scale.
- Devices are fabricated layer-by-layer using precise patterning techniques.
- Fabrication is carried out in cleanroom environments to avoid contamination.
- It is the foundation of modern VLSI and semiconductor manufacturing.
- Fabrication is a **repetitive cycle of patterning + modification + deposition**.
- Precision at nanometer scale is critical for device performance.
- Alignment between layers is crucial for circuit functionality.
- Process variations can significantly impact device characteristics.
- Yield (percentage of working chips) is a key manufacturing metric.

### 1.17.1 Key Process Steps

- **Oxidation:** Growth of silicon dioxide ( $\text{SiO}_2$ ) layer for insulation and protection.
- **Photolithography:** Pattern transfer using light, masks, and photoresist.
- **Etching:** Removal of material (wet or dry) to define structures.
- **Doping (Diffusion / Ion Implantation):** Introduction of impurities to form p-type or n-type regions.
- **Deposition:** Addition of material layers (chemical or physical).

### 1.17.2 Supporting Processes

- **Annealing:** Heat treatment to repair damage and activate dopants.
- **Planarization (CMP):** Smoothing wafer surface for uniform layers.
- **Metallization:** Formation of metal interconnects for electrical connections.
- **Passivation:** Protective layer to prevent contamination and damage.

### 1.17.3 Challenges

- Scaling down device size increases fabrication complexity.
  - Contamination can destroy devices  $\rightarrow$  strict cleanroom control required.
  - Process variations affect performance and reliability.
  - High cost of fabrication facilities (fabs).
  - Thermal and material limits at nanoscale.
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## 1.18 Books and references

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